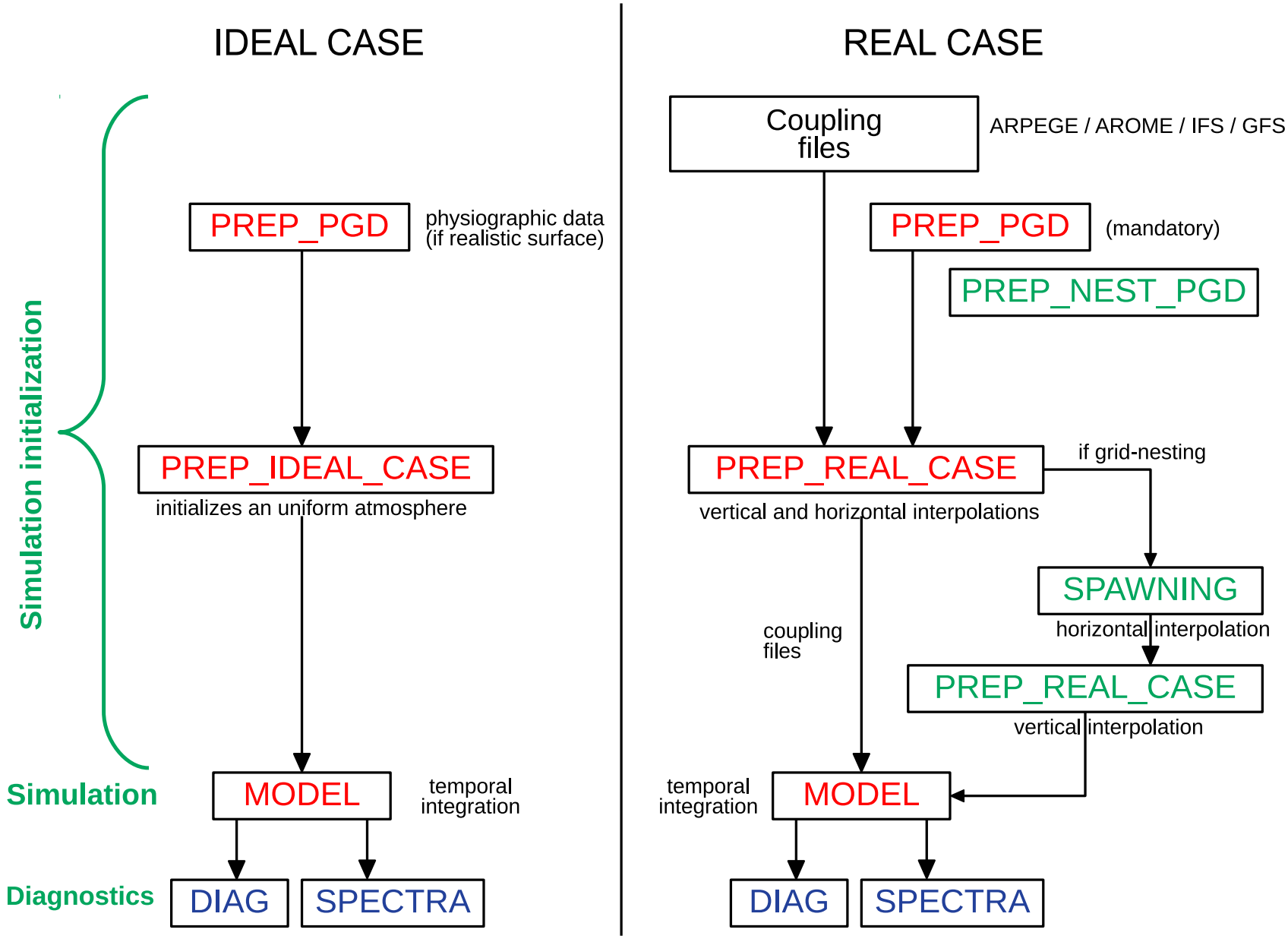
Real case

MesoNH Tutorial - 2-5 November 2026


IDEAL CASE
REAL CASE
Simulation initialization
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
DIAG
SPECTRA
Coupling
files
ARPEGE / AROME / IFS / GFS
PREP_PGD
(mandatory)
PREP_NEST_PGD
PREP_REAL_CASE
vertical and horizontal interpolations
if grid-nesting
SPAWNING
horizontal interpolation
coupling
files
PREP_REAL_CASE
vertical interpolation
temporal
integration
MODEL
DIAG
SPECTRA

One-domain simulation

Creation of PGD file

Program PREP_PGD

interpolation on horizontal grid of input fields
Output PGD : file with projection, domain and 2D fields

Input files for :

- orography
- cover
- sand fraction
- clay fraction

External data

Post-process and vizualize

DOCUMENTATION

Scientific & References

User's guide

Tutorial material

Developer's guide

APPLICATIONS

Namelist catalogue

Aerosols

Chemistry

Boundary-layer LES

Electricity

Wildland fire

Wind turbine

Ocean-wave-atmosphere

COMMUNITY

Meetings

Publications

Newsletters

Contributors

EXTERNAL RESSOURCES

Data

Physiographic invariant surface data

Initialization of atmospheric and surface data

Tools

External data

Note

Depending of your application:

- Physiographic static surface data are needed by the PREP_PGD program.
- Atmospheric and surface data are needed by the PREP_REAL_CASE program.

- Physiographic invariant surface data
  - Download
  - Topography
  - Soil texture
  - Land use
  - Additional databases
- Initialization of atmospheric and surface data
  - Extract ECMWF data
  - Extract Meteo-France data
  - Extract GFS data
  - Extract HRRR data
  - Extract ICON data
  - Extract CAMS data

Previous Next

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Namelist PRE_PGD1.nam

```
&NAM_PGDFILE CPGDFILE='PGD_D1' /PGD file name

&NAM_CONF_PROJ XLAT0=37., XLON0=5.1, lat/lon reference
      cone factor  XRPK=0.58, XBETA=0 /rotation angle

&NAM_CONF_PROJ_GRID XLATCEN=38., XLONCEN=5., center lat/lon
                    NIMAX=12, NJMAX=10, number of points in I and J
                    XDX=2000., XDY=2000. /ΔX and ΔY

&NAM_PGD_SCHEMES CNATURE='ISBA', CSEA='SEAFLX',
surface schemes        CWATER='WATFLX', CTOWN='TEB' /

&NAM_COVER YCOVER='ecoclimats_v2'/surface cover database
&NAM_ZS YZS='gtopo30'/orography database
&NAM_ISBA YCLAY='clay_fao',YSAND='sand_fao'/clay and sand fractions
```

NIMAX and NJMAX must be equal to $2^n\ 3^m\ 5^p$

Atmosphere and surface fields

Data are from

- a large-scale model (GRIB)

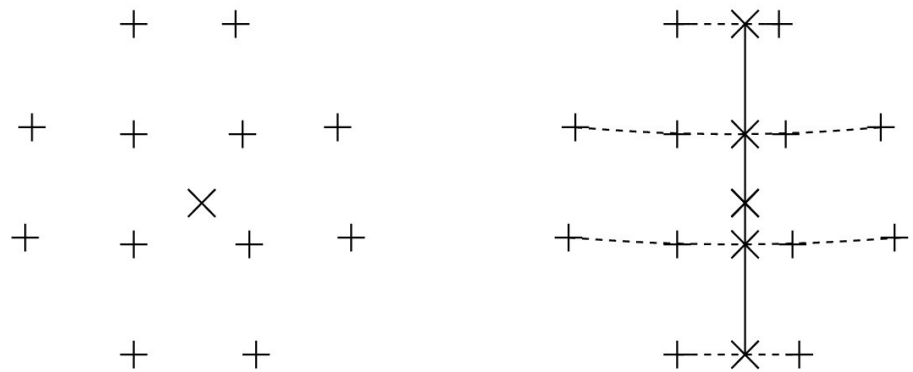
Model	**Access**	**Period**	**Area**
IFS, ERA5	Open*	1970 -	Global
ARPEGE	Not open**	1998 -	Global
GFS	Open	2006 -	Global
ICON	Open	Real-time	Global + Europe
AROME	Not open**	2008 -	France metropoli.
AROME-OM	Not open**	2016 -	France Outre-mer
HRRR	Open	2014 -	USA

- or another MesoNH simulation (netCDF)

* unless real-time ** except for MF people ; otherwise a sampling is possible

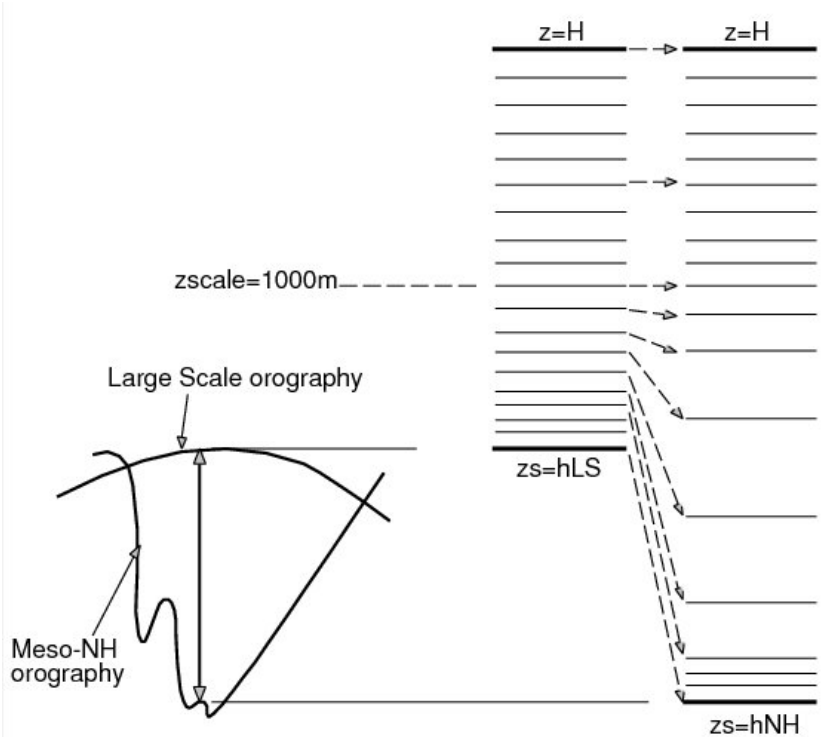
PREP_REAL_CASE

- Horizontal interpolation at each level
  of U,V,T,q,Ps, surface 2D fields on the PGD grid from the nearest 12 points.



PREP_REAL_CASE

- Vertical interpolation



déplacement vertical selon la fonction 'shift'

entre hLS et zscale

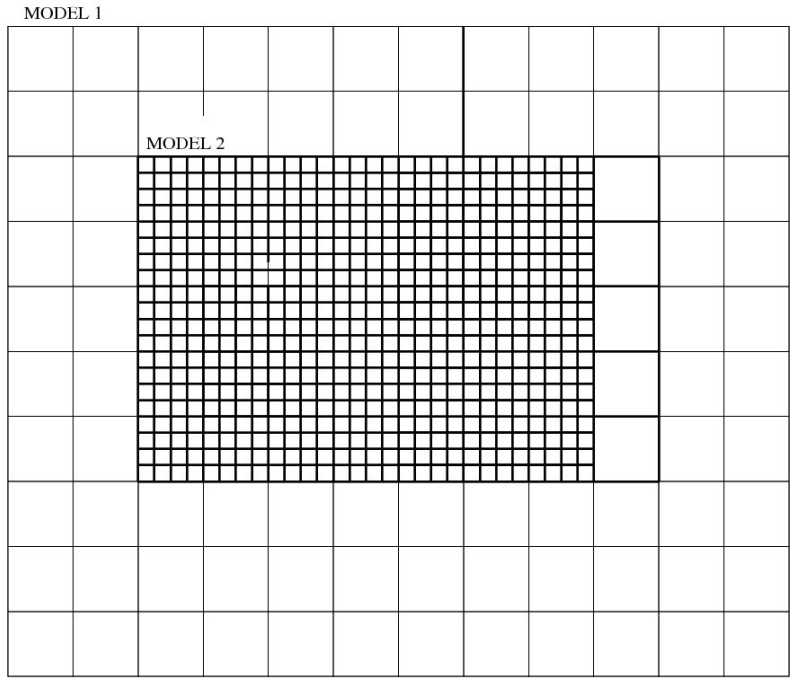
Namelist PRE_REAL1.nam

It must be made for the initial file and **all** coupling files

```
&NAM_FILE_NAMES
      HATMFILE ='arpege.FC.20110128.21' ,atmospherical file
      HATMFILETYPE='GRIBEX',type of atmospherical file
      HPGDFILE ='PGD_D1' ,PGD file name
      CINIFILE='PREP_28JAN21H_D1' /name of output file

&NAM_VER_GRID NKMAX=120,number of points in Z
              YZGRID_TYPE='FUNCTN',FUNCTN or MANUAL
              ZDZGRD=10., ZDZTOP=500.,ΔZ at ground/at top
              ZZMAX_STRGRD=2500., Height for streching change
              ZSTRGRD=5.,ZSTRTOP=7. /streching at ground/at top
```

two-domain simulation



Grid-nesting : horizontal grids

- a PGD file for every model
- All PGDs must have the same averaged orography over their common area : PREP_NEST_PGD

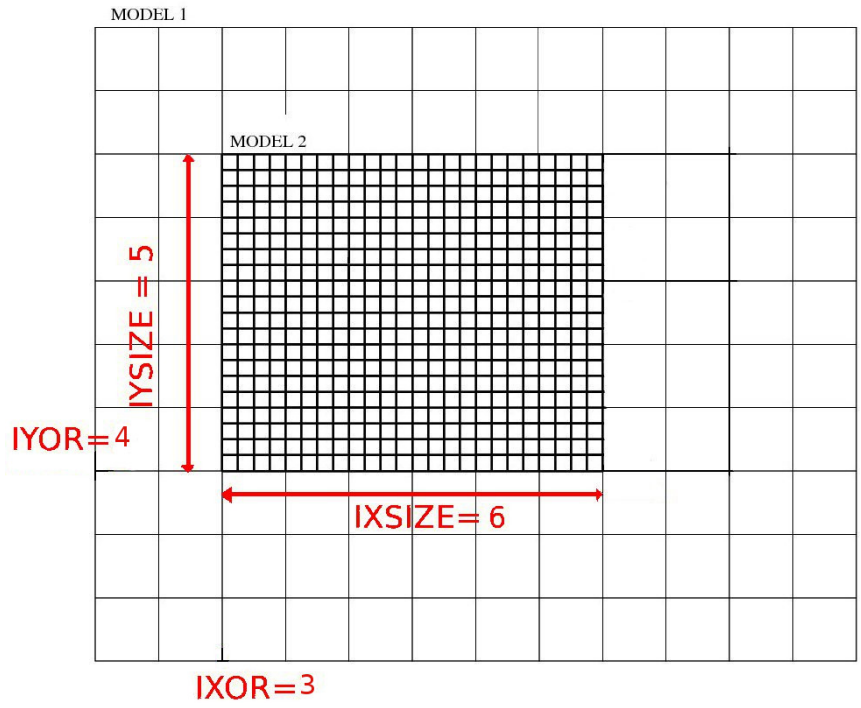
ALL PGD FILES MUST BE MADE AND "NESTED" BEFORE THE SIMULATION

Grid-nesting : initial and coupling files

- prepare initial + coupling file for the parent domain with PREP_REAL_CASE
- prepare initial file for the child domain(s)
  - SPAWNING : horizontal interpolation of 3D fields from parent's model to child's model
  - PREP_REAL_CASE :vertical interpolation (parent to child)
- a file EXSEGn.nam for every domain

child's PGD : PREP_PGD

- PRE_PGD1.nam :




```
&NAM_PGDFILE
    CPGDFILE='PGD_D2' /

&NAM_CONF_PROJ /

&NAM_CONF_PROJ_GRID /

&NAM_PGD_GRID
    YINIFILE='PGD_D1',
    YINIFILETYPE='MESONH'
/
&NAM_INIFILE_CONF_PROJ IXOR=3, IYOR=4,
                       IXSIZE=6, IYSIZE=5,
                       IDXRATIO=4, IDYRATIO=4 /

&NAM_COVER YCOVER='ecoclimats_v2' /
&NAM_ZS YZS='gtopo30' /
&NAM_ISBA YCLAY='clay_fao', YSAND='sand_fao' /
```

PREP_NEST_PGD

```
&NAM_PGD1 YPGD1= 'PGD_D1' /parent PGD file
&NAM_PGD2 YPGD2= 'PGD_D2' , IDAD = 1 /
                  First child PGD file / number of the parent file
&NAM_PGD3 /
...
&NAM_PGD8 /
&NAM_NEST_PGD YNEST= 'e1' /string of 2 characters to be added to the PGD file names
to define the corresponding output PGD file names
```

Preparing initial file for D2 : SPAWNING

```
&NAM_GRID2_SPA /
&NAM_LUNIT2_SPA CINIFILE = 'PREP_28JAN21H_D1' ,
                name of the initial file
                 CINIFILEPGD = 'PGD_D1.neste1',
                PGD file associated to CINIFILE
                 YDOMAIN = 'PGD_D2.neste1',
                PGD file name (output domain)
```

The SPAWNING stage must be followed by **PREP_REAL_CASE** if the domain is not flat

new file produced : PREP_28JAN21H_D1.spa04.nc

PREP_REAL_CASE

PRE_REAL1.nam :

```
&NAM_FILE_NAMES
      HATMFILE ='PREP_28JAN21H_D1.spa04',input file (spawned file)
      HATMFILETYPE='MESONH',
      HPGDFILE ='PGD_D2.neste1',
      CINIFILE ='PREP_28JAN21H_D2' /output name of prep_real_case

&NAM_REAL_CONF NVERB=5 /

&NAM_VER_GRID YZGRID_TYPE='SAMEGR' /

&NAM_PREP_SURF_ATM          namelist from SURFEX
       CFILE = 'PREP_28JAN21H_D1',
       CFILETYPE = 'MESONH' ,
       CFILEPGD="PGD_D1.neste1",
       CFILEPGDTYPE = 'MESONH' /
```

Namelist

Principe

One namelist EXSEG**n**.nam for each model

Model 1 (dad) : EXSEG1.nam

Namelist ended by **n** are relative to model 1
Other namelists are common for all models

Modele 2 (son) : EXSEG2.nam

There is only an initial file (no coupling file)
Only namelists ended by **n** are taken into account

MESONH with grid-nesting

file EXSEG1.nam

```
&NAM_LUNITn CINIFILE = "PREP_28JAN21H_D1",
            CINIFILEPGD = "PGD_D1.neste1"/
            CCPLFILE(1) = "PREP_29JAN00H_D1"/

&NAM_DYNn XTSTEP = 60., CPRESOPT = "ZRESI",
          NITR=8, LHORELAX_UVWTH = T,
          LHORELAX_RV = T, LVE_RELAX = T,
          NRIMX = 5, NRIMY = 5, XRIMKMAX = 0.0083 /

&NAM_ADVn CUVW_ADV_SCHEME = "WENO_K",
           NWENO_ORDER=5 CTEMP_SCHEME='RK53',
           CMET_ADV_SCHEME = "PPM_01" /

&NAM_PARAMn CTURB = "TKEL", CRAD = "ECRA",
            CSCONV = "EDKF", CDCONV = "KAFR",
            CCLOUD = "ICE3"/
```

```
&NAM_PARAM_RADn XDTRAD = 3600.,
                XDTRAD_CLONLY = 3600.,
                NRAD_COLNBR = 400 /

&NAM_PARAM_KAFRn XDTCONV = 300., NICE = 1,
                 LREFRESH_ALL = T, LDOWN = T /

&NAM_LBCn CLBCX = 2*"OPEN", CLBCY = 2*"OPEN" /

&NAM_TURBn CTURBLEN = "BL89",
           CTURBDIM = "1DIM"

&NAM_CONF CCONF = "START", start/restart simulation
          NMODEL = 2, number of [parent + child(s)] models
          CEXP = "CTRL0", CSEG = "SEG01" /

          name of the outputs = CEXP.Nmodel.CSEG.00(n)

&NAM_DYN XSEGLEN = 400., LCORIO = T,
         XALKTOP = 0.001, XALZBOT = 14500. /
```

```
&NAM_NESTING NDAD(2) = 1,  1 is the parent of the child number 2
             NDTRATIO(2) = 4,  ratio of the timestep for the child number 2
               XWAY(2) = 2.  1 = one-way; 2 = two-way interactions /

&NAM_BACKUP XBAK_TIME(1,1)=100/
```

file EXSEG2.nam

```
&NAM_LUNITn CINIFILE = "PREP_28JAN21H_D2"/
            CINIFILEPGD = "PGD_D2.neste1"/

&NAM_DYNn CPRESOPT = "ZRESI",
          LVE_RELAX = T,NITR=8

&NAM_ADVn CUVW_ADV_SCHEME = "WENO_K",
          CMET_ADV_SCHEME = "PPM_01" /

&NAM_PARAMn CTURB = "TKEL", CRAD = "ECRA",
            CSCONV = "NONE", CDCONV = "NONE",
            CCLOUD = "ICE3"/
```

```
&NAM_PARAM_RADn XDTRAD = 1800.,
                XDTRAD_CLONLY = 1800.,
                LCLEAR_SKY = F, NRAD_COLNBR = 400 /

&NAM_PARAM_KAFRn XDTCONV = 300., NICE = 1,
                 LREFRESH_ALL = T, LDOWN = T /

&NAM_LBCn CLBCX = 2*"OPEN", CLBCY = 2*"OPEN",
          XCPHASE = 20. /

&NAM_TURBn XIMPL = 1., CTURBLEN = "DEAR",
           CTURBDIM = "3DIM"
```